Drywall & Metal Stud Takeoff Cheat Sheet

Sheet counts, waste, studs and track, gauges, rated walls, and which plan sheets to pull. Everything here is a standard rule of thumb or plain arithmetic. Check each one against your own plans, specs and job history.

1. Drywall sheets from wall length

Board area = wall LF x height x sides x layers. Ignore small openings (the cut-out usually becomes waste anyway); subtract only big ones such as storefronts and wide cased openings. Then divide by the sheet area and round up **once per board type**, after waste.

Wall height	SF per LF (1 side, 1 layer)	4x8 sheets per 100 LF (before waste)	4x12 sheets per 100 LF (before waste)
8 ft	8	25.0	16.7
9 ft	9	28.1	18.8
10 ft	10	31.3	20.8
12 ft	12	37.5	25.0

Sheet areas: 4x8 = 32 SF, 4x9 = 36 SF, 4x10 = 40 SF, 4x12 = 48 SF. Both sides of a partition = double the figures above. Two layers = double again. Exterior walls usually get board on the inside face only.

Worked example. 40 LF of wall, 9 ft high, one layer, one side: 40 x 9 = 360 SF. With 10% waste, 396 SF. On 4x8 sheets that is 12.4, so 13 sheets.

Ceilings: hard-lid area from the reflected ceiling plan, same sheet math. Soffits and bulkheads are the easy ones to miss.

2. Waste factors (starting points, not a standard)

Job type	Common starting range	Why
Long straight walls, corridors, 4x12 board	about 10%	Few offcuts, few end joints
Apartments, small rooms, closets, many corners	about 12 to 15%	More cuts that cannot be reused
Ceilings with lights, diffusers, access panels, soffits	toward the high end	Cut-outs and odd shapes

- Total board area per type across the job first, then apply waste once. Rounding up room by room over-orders.
- Keep 5/8 in. Type X, moisture-resistant and standard board as separate lines.
- Your own number is better: sheets delivered minus sheets hung, on your last few jobs.
- Do not use waste to cover a bad takeoff. Fix the area first.

3. Metal studs and track per LF

Studs = (wall length in inches / spacing) + 1. A 20 ft wall at 16 in. o.c.: 240 / 16 + 1 = 16 studs. For totals across many walls, use the per-LF rule below and add extras.

Spacing (o.c.)	Studs per LF	Studs per 100 LF (base)	Track per 100 LF
12 in.	1.00	100	200 LF + laps
16 in.	0.75	75	200 LF + laps
24 in.	0.50	50	200 LF + laps

- **Track:** top and bottom, so wall LF x 2. Add about 5 to 10% for laps and cuts, then divide by stick length (commonly 10 ft).
- **Extras (typical, check the details):** 1 to 2 studs per corner or intersection; about 2 per door or window for jambs, plus header and cripples as drawn; 1 extra where a wall dead-ends.
- **Stud length:** order to the wall height in the wall-type detail, to ceiling or to deck. Deck-height walls are a common undercount.
- **Deflection or slip track** at the head of walls that run to the deck, and **backing** for cabinets, grab bars, handrails and wall-hung fixtures (often only shown on interior elevations).

Worked example. 300 LF of 3-5/8 in. partition at 16 in. o.c., 10 ft tall, 8 doors, 14 corners/intersections: base 225 studs, doors 16, corners 28, about 269 studs at 10 ft. Track: 600 LF x 1.07 = about 642 LF, or 65 sticks of 10 ft.

4. Gauge notes

Gauge	Mils	Typical use
25 ga	18 mil	Ordinary non-structural interior partitions
20 ga drywall	30 mil	Taller or heavier-finish non-structural walls
20 ga structural	33 mil	Load-bearing and exterior wall framing (the engineer sets it)
18 ga	43 mil	Structural, shorter walls
16 ga	54 mil	Taller structural walls, headers, jambs
14 ga / 12 ga	68 / 97 mil	Heavy structural

- 1 mil = 0.001 in. A lower gauge number means thicker steel. Stud gauge is not sheet-metal gauge.
- Part number example: 362S162-43 = 3-5/8 in. web, 1-5/8 in. flange, 43 mil.
- If the plans do not state a gauge, **do not assume one silently.** Write it down as an assumption or ask. Max wall height comes from the manufacturer's limiting-height tables, not a rule of thumb.

5. Fire-rated walls

- Count rated assemblies as their own wall types. Different board, more layers, perimeter sealant.
- Rated walls commonly use 5/8 in. Type X, often one or two layers per face. **The tested design listed in the plans governs.** Count each layer as its own SF.
- Rated walls usually run to the deck. Using ceiling height undercounts both board and studs. Example: 120 LF at 12 ft to deck, one layer each face = 120 x 12 x 2 = 2,880 SF; at a 9 ft ceiling it would show 2,160 SF.
- Stud gauge and spacing are set by the assembly. Do not substitute.
- Penetration firestopping is usually another trade's scope. Confirm it. Price rated walls on their own line.

6. Finishing and fasteners (rough ratios)

- Joint tape: about 370 to 400 LF of paper tape per 1,000 SF of board.
- Screws: about one per SF of board as a starting point. Ceilings and rated assemblies are tighter; two-layer walls need longer screws on the face layer.
- Joint compound: use the manufacturer's coverage chart and the level of finish in the spec (Level 0 to 5 under GA-214). Not one fixed number.
- Control joints: ASTM C840 sets maximum spacing, including about 30 ft along a wall run. Long corridors and big ceilings need them even if not drawn.
- Bead and trim: outside corner bead by LF of outside corners (include soffits and column wraps); J-bead or L-bead where board stops against other materials.

7. Which plan sheets to pull (checklist)

	Plan sheet or document	What you pull from it
[ ]	Floor plans (every floor)	Wall LF by wall-type tag, doors, openings
[ ]	Wall-type legend / partition schedule	Stud size, gauge, spacing, board, layers, insulation, height. Check the legend is the one for that sheet.
[ ]	Enlarged plans and details	Walls only shown at larger scale
[ ]	Building and wall sections	Wall height: to ceiling or to deck; head-of-wall details
[ ]	Reflected ceiling plans	Hard-lid area, soffits, bulkheads, ceiling heights
[ ]	Life safety / rated wall plan	Rated corridors, shafts, demising walls
[ ]	Interior elevations	Backing and blocking for fixtures
[ ]	Door and window schedules	Opening sizes and counts
[ ]	Finish schedule and specs (Div 09)	Board type, level of finish, framing requirements
[ ]	Structural sheets and general notes	Gauge, loads and notes that do not show on the wall-type sheet
[ ]	Addenda and revision clouds	Anything that changed since the set you started from

Before you send the bid

- Recount a few walls by hand against the totals. If anything is off, widen the recount.
- Write down assumptions and exclusions (gauge, finish level, who does firestopping).
- Keep totals split by wall type so you can trace any number back to the wall it came from.

Skip the hand count. ReadMyBlueprint reads the wall-type legend on a digital plan set, matches each wall to its type, and flags the walls it is unsure about for you to confirm. Your first plan set is free, no card needed.
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